

Practical Session 5

Perform packet tracing and determine the content of the packet.

Practical Title: Working with Wireshark (Packet Analysis)

Problem statement

Perform packet tracking and determine the content of the packet.

Instructions

- Work with Wireshark.
- Start capturing packets.
- Analyse the packets and get the required details.

Objective: By the end of this practical, you will be able to analyse network traffic successfully.

Software/s required, if any:

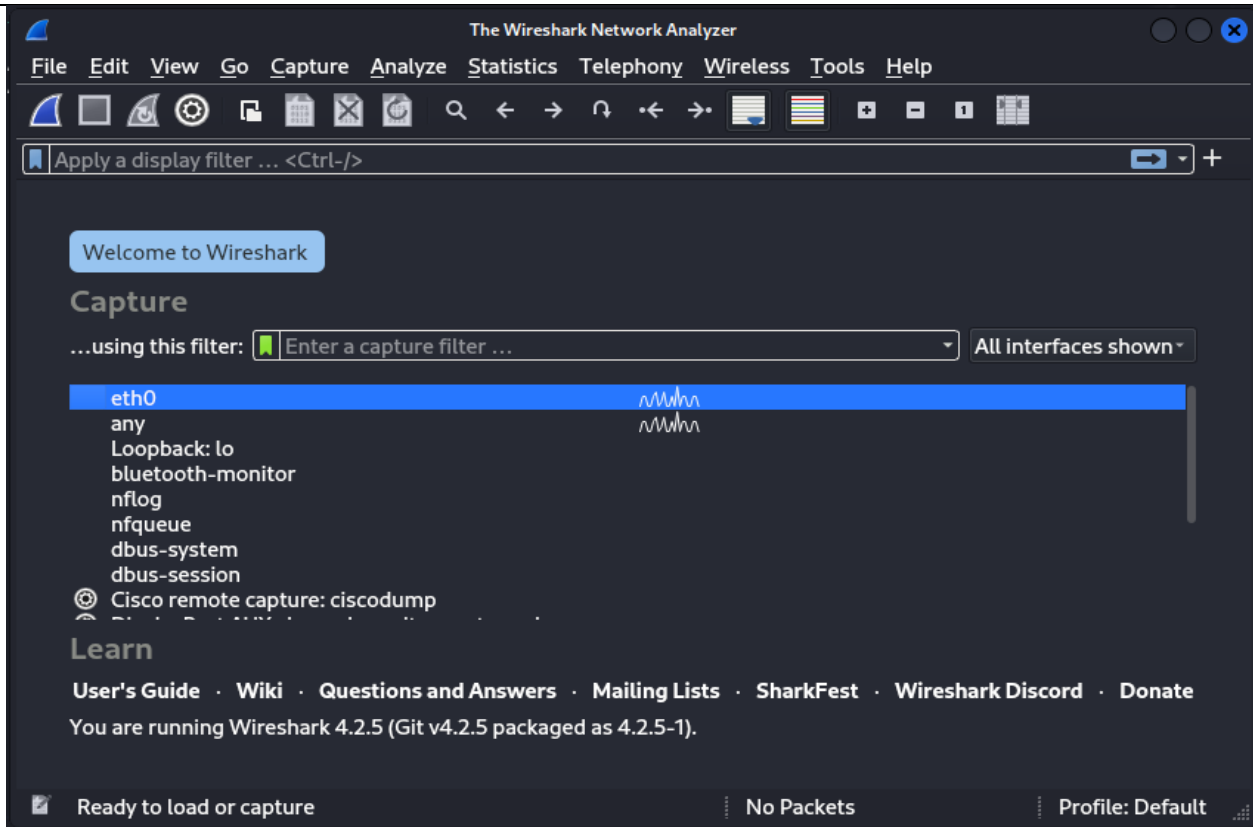
1. Wireshark

Hardware required, if any: NA

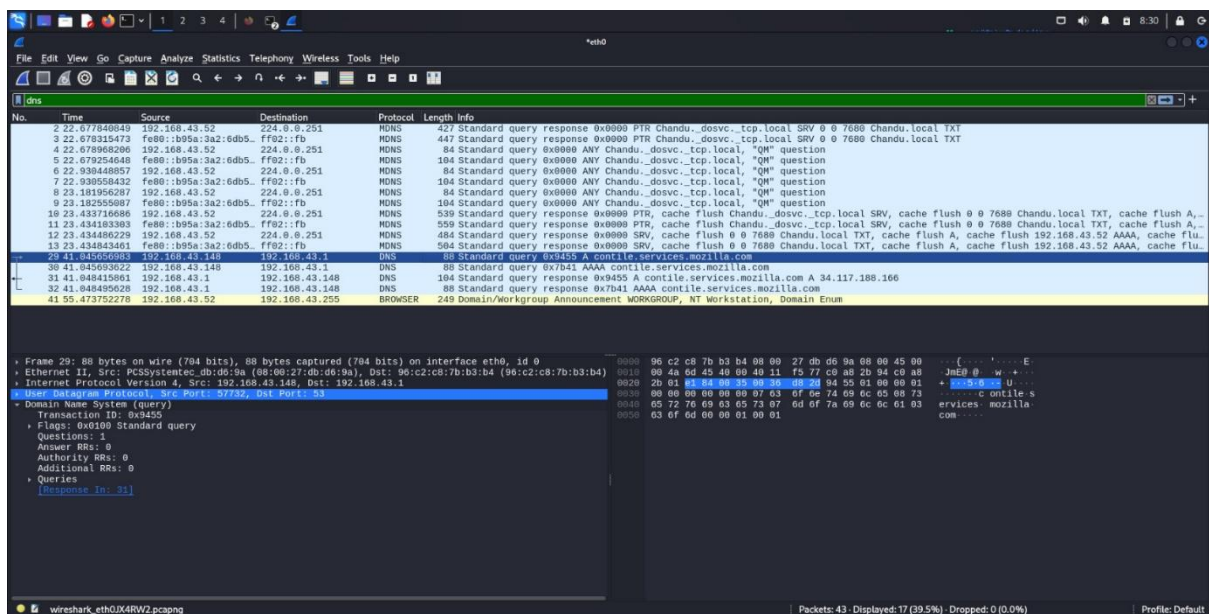
External references, any : No

Information to be filled in by learner

Whireshark



1. dns



2. arp

Wireshark interface showing ARP traffic on interface eth0. The packet list displays several ARP requests and responses. The selected packet (No. 1) is an ARP request from 96:c2:c8:7b:b3:b4 to the broadcast address ff:ff:ff:ff:ff:ff.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.52? Tell 192.168.43.1
2	20.233966498	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.52? Tell 192.168.43.1
3	40.212955425	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.52? Tell 192.168.43.1
4	53.774672403	PCSSystemtec_db:d6...	Broadcast	ARP	42	Who has 192.168.43.1? Tell 192.168.43.148
5	53.783567613	96:c2:c8:7b:b3:b4	PCSSystemtec_db:d6...	ARP	60	192.168.43.1 is at 96:c2:c8:7b:b3:b4
6	58.909566642	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.148? Tell 192.168.43.1
7	58.909662765	PCSSystemtec_db:d6...	96:c2:c8:7b:b3:b4	ARP	42	192.168.43.148 is at 08:00:27:db:d6:9a
8	60.78.178824384	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.52? Tell 192.168.43.1

Packet details for No. 1: Ethernet II, Src: 96:c2:c8:7b:b3:b4 (96:c2:c8:7b:b3:b4), Dst: Broadcast (ff:ff:ff:ff:ff:ff). Address Resolution Protocol (request). Hardware type: Ethernet (1). Protocol type: IPv4 (0x0008). Hardware size: 6. Protocol size: 4. Opcode: request (1). Sender MAC address: 96:c2:c8:7b:b3:b4 (96:c2:c8:7b:b3:b4). Sender IP address: 192.168.43.1. Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00). Target IP address: 192.168.43.52.

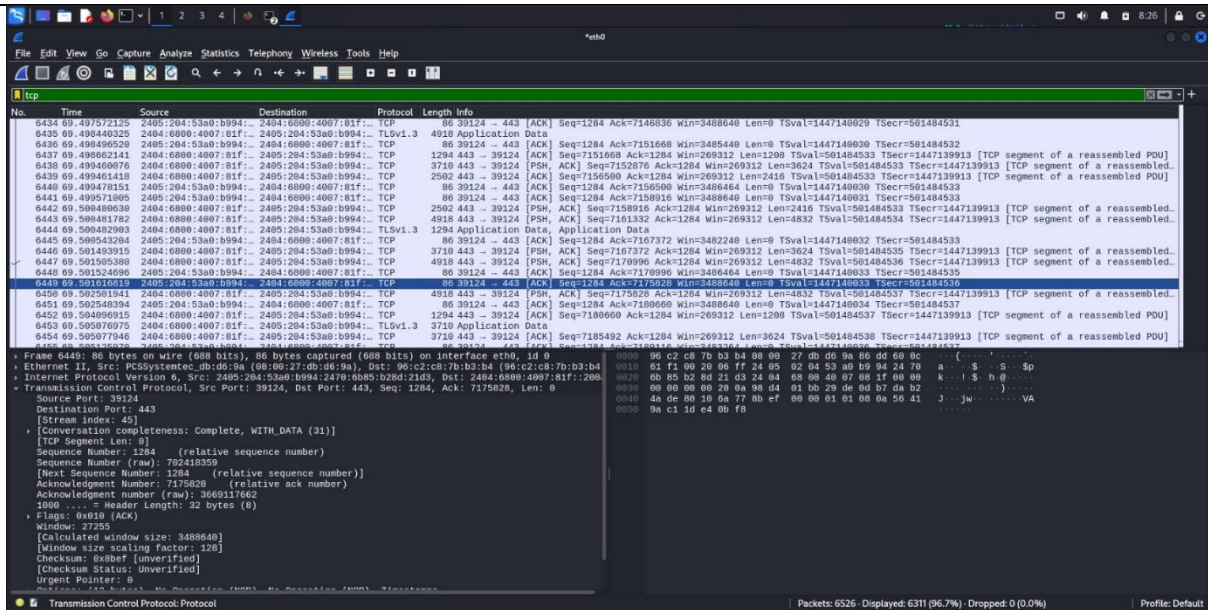
3. http

Wireshark interface showing HTTP traffic on interface eth0. The packet list displays a GET request and its response. The selected packet (No. 1) is a GET request for / HTTP/1.1.

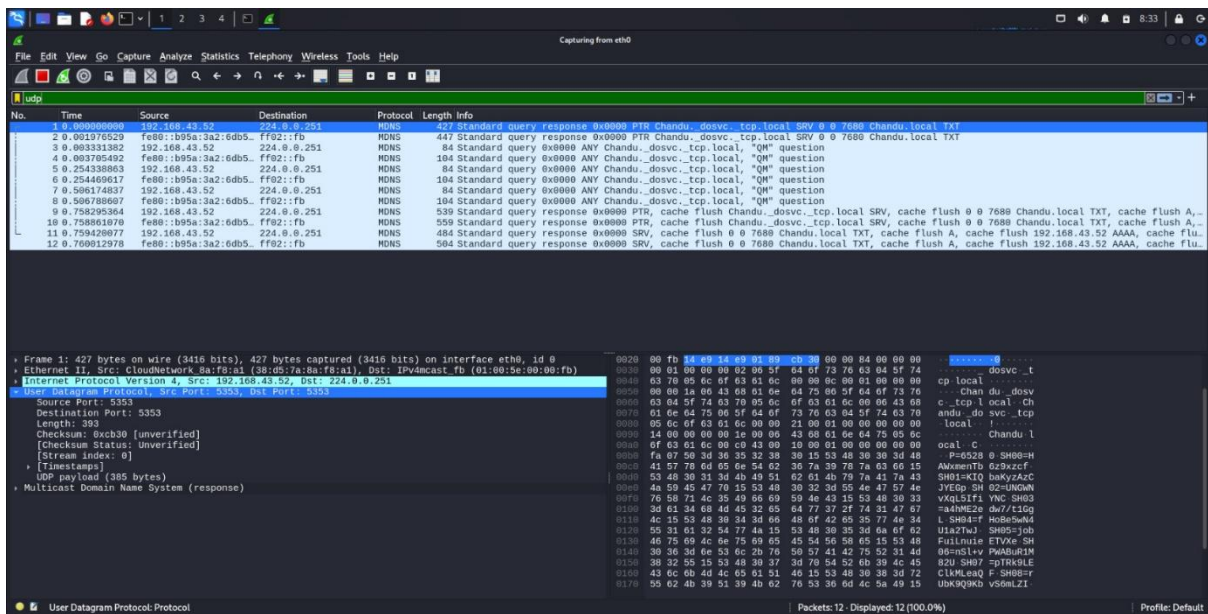
No.	Time	Source	Destination	Protocol	Length	Info
26	54.158475641	2405:204:53a0:b994::	2405:204:1606:1731::	OCSP	502	Request
28	54.206254736	2405:204:1606:1731::	2405:204:53a0:b994::	OCSP	575	Response
52	54.61162254	192.168.43.190	192.168.43.148	HTTP	1034	HTTP/1.1 200 OK (text/html)
98	54.98829467	192.168.43.148	34.107.221.82	HTTP	381	GET /success.txt?ip=4 HTTP/1.1
93	54.99330338	2405:204:53a0:b994::	2600:1901:0:38d7::	HTTP	381	GET /success.txt?ip=6 HTTP/1.1
95	55.055613828	34.107.221.82	192.168.43.148	HTTP	282	HTTP/1.1 200 OK (text/plain)
96	55.055615721	2600:1901:0:38d7::	2405:204:53a0:b994::	HTTP	302	HTTP/1.1 200 OK (text/plain)
115	55.368354175	192.168.43.148	192.168.43.190	HTTP	349	GET /index.css HTTP/1.1
116	55.368369472	192.168.43.190	192.168.43.148	HTTP	1188	HTTP/1.1 200 OK (text/css)
118	55.37637334	192.168.43.148	192.168.43.190	HTTP	338	GET /jquery.min.js HTTP/1.1
122	55.371849178	192.168.43.148	192.168.43.190	HTTP	344	GET /animatedcolllapse.js HTTP/1.1
126	55.373399508	192.168.43.190	192.168.43.148	HTTP	142	HTTP/1.1 200 OK (application/javascript)
132	55.374610881	192.168.43.190	192.168.43.148	HTTP	7314	HTTP/1.1 200 OK (application/javascript)
134	55.374720808	192.168.43.148	192.168.43.190	HTTP	363	GET /images/owasp.png HTTP/1.1
135	55.374843743	192.168.43.148	192.168.43.190	HTTP	366	GET /images/mandiant.png HTTP/1.1
142	55.375226063	192.168.43.148	192.168.43.190	HTTP	366	GET /images/knob-add.png HTTP/1.1
144	55.375606036	192.168.43.148	192.168.43.190	HTTP	349	GET /index.css HTTP/1.1
160	55.376783148	192.168.43.190	192.168.43.148	HTTP	1189	HTTP/1.1 200 OK (text/css)
168	55.376853178	192.168.43.190	192.168.43.148	HTTP	1470	HTTP/1.1 200 OK (PNG)
169	55.380579727	192.168.43.190	192.168.43.148	HTTP	2406	HTTP/1.1 200 OK (PNG)

Packet details for No. 1: Ethernet II, Src: PCSSystemtec_db:d6:9a (08:00:27:db:d6:9a), Dst: PCSSystemtec_db:ib:c2 (08:00:27:ib:c2). Internet Protocol Version 4, Src: 192.168.43.148, Dst: 192.168.43.190. Transmission Control Protocol, Src Port: 48982, Dst Port: 80, Seq: 1, Ack: 1, Len: 338. Hypertext Transfer Protocol. GET / HTTP/1.1. Host: 192.168.43.190. User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0. Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8. Accept-Language: en-us,en;q=0.5. Accept-Encoding: gzip, deflate. Connection: keep-alive. Upgrade-Insecure-Requests: 1. [HTTP request 1/4]. [Response in frame 5]. [Next request in frame 115].

4. tcp



5. udp



6. tcp==80

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp==80

No.	Time	Source	Destination	Protocol	Length	Info
80	53.792255049	2405:204:53a0:b994::	2600:1901:0:92a9::	TCP	86	34752 → 443 [ACK] Seq=47 Win=249 Len=0 TSval=596555481 TSecr=641791535
81	53.798197178	34.117.188.166	192.168.43.148	TLSv1.2	105	Application Data
82	53.799218707	192.168.43.148	34.117.188.166	TCP	66	40990 → 443 [ACK] Seq=40 Ack=40 Win=249 Len=0 TSval=2174548931 TSecr=1455942640
83	53.842895593	2405:204:53a0:b994::	2405:200:1608:1731::	TCP	86	[TCP Previous segment not captured] 40024 → 80 [FIN, ACK] Seq=2 Ack=1 Win=249 Len=0 TSval=3354250916 TSecr=3883687579
84	53.842941116	2405:204:53a0:b994::	2405:200:1608:1731::	TCP	86	[TCP Previous segment not captured] 40024 → 80 [FIN, ACK] Seq=2 Ack=1 Win=249 Len=0 TSval=3354250916 TSecr=3883687579
85	53.850603557	2405:200:1608:1731::	2405:204:53a0:b994::	TCP	86	80 → 40036 [FIN, ACK] Seq=1 Ack=3 Win=501 Len=0 TSval=3303690139 TSecr=3354250916
86	53.886084729	2405:200:1608:1731::	2405:204:53a0:b994::	TCP	86	80 → 40024 [FIN, ACK] Seq=1 Ack=3 Win=501 Len=0 TSval=3803690139 TSecr=3354250916
87	53.886132493	2405:204:53a0:b994::	2405:200:1608:1731::	TCP	86	40036 → 80 [ACK] Seq=3 Ack=2 Win=249 Len=0 TSval=3354250959 TSecr=3883690139
88	53.886156083	2405:204:53a0:b994::	2405:200:1608:1731::	TCP	86	40024 → 80 [ACK] Seq=3 Ack=2 Win=249 Len=0 TSval=3354250959 TSecr=3883690139
89	54.073449991	192.168.43.148	34.149.188.289	TLSv1.2	105	Application Data
90	54.073584867	192.168.43.148	34.187.243.93	TLSv1.2	105	Application Data
91	54.756180984	34.107.243.93	192.168.43.148	TLSv1.2	105	Application Data
92	54.756182495	34.149.188.289	192.168.43.148	TLSv1.2	105	Application Data
93	54.756210112	192.168.43.148	34.107.243.93	TCP	66	50150 → 443 [ACK] Seq=40 Ack=40 Win=249 Len=0 TSval=3326713995 TSecr=2404470652
94	54.756244328	192.168.43.148	34.149.188.289	TCP	66	50150 → 443 [ACK] Seq=40 Ack=40 Win=249 Len=0 TSval=3357151242 TSecr=3205445210
95	56.675545277	2405:204:53a0:b994::	2600:1901:0:9e3f::	TLSv1.2	125	Application Data
96	56.763962173	2600:1901:0:9e3f::	2405:204:53a0:b994::	TLSv1.2	125	Application Data
97	56.763928258	2405:204:53a0:b994::	2600:1901:0:9e3f::	TCP	86	50050 → 443 [ACK] Seq=40 Ack=40 Win=249 Len=0 TSval=2550195393 TSecr=3830765875
98	58.909708705	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.148? Tell 192.168.43.1
99	58.909726759	PCSSystemec.db:d6::	96:c2:c8:7b:b3:b4	ARP	42	192.168.43.148 is at 00:00:27:db:d6:0a
100	61.214881972	96:c2:c8:7b:b3:b4	Broadcast	ARP	60	Who has 192.168.43.52? Tell 192.168.43.1

Frame 93: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemec.db:08:0a:00:00:27:db:d6:0a, Dst: 96:c2:c8:7b:b3:b4 (96:c2:c8:7b:b3:b4)

Internet Protocol Version 4, Src: 192.168.43.148, Dst: 34.197.243.93

Transmission Control Protocol, Src Port: 59410, Dst Port: 443, Seq: 40, Ack: 40, Len: 0

Source Port: 59410

Destination Port: 443

[Stream index: 7]

[Conversation completeness: Incomplete (12)]

[TCP Segment Len: 0]

Sequence Number: 40 (relative sequence number)

Sequence Number (raw): 1759729182

[Next Sequence Number: 40 (relative sequence number)]

Acknowledgment Number: 40 (relative ack number)

Acknowledgment number (raw): 3663807016

1000 = Header Length: 32 bytes (8)

Flags: 0x010 (ACK)

Window: 249

[Calculated window size: 249]

Window size scaling factor: -1 (unknown)

Checksum: 0x922c (unverified)

[Checksum Status: Unverified]

Urgent Pointer: 0

eth0: <live capture in progress>

Packets: 100 · Displayed: 100 (100.0%)

Profile: Default